

Smart Metro Ticket Booking System

IDEATION PHASE

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	2 Marks

1. Problem Statement

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1.1 Background

Metro rail remains one of the most heavily used forms of urban public transport, prized for its speed, affordability, and its role in easing road congestion. Even so, ticket purchase in many metro networks still relies on manual counters, coin/token machines, or kiosks that are only partly digitized.

To travel, a commuter must go to the station, join a queue, pick the origin and destination stops, settle the fare in cash or by card, and collect a paper ticket or token before being let onto the platform.

1.2 Problem Statement

"Passengers using the metro regularly encounter lengthy queues, fare-calculation mistakes made by hand, restricted counter hours, and reliance on physical tokens or tickets. Together these cause delays, crowded ticket counters, and a frustrating travel experience — particularly during rush hour and for people travelling for the first time or with disabilities."

1.3 Problem Statement Canvas (Who / What / Why)

Element	Description
I am (User)	A regular office-goer, student, occasional traveller, senior citizen, or a differently-abled rider who depends on the metro.
Trying to	Purchase a metro ticket quickly between a chosen source and destination without queueing.
But	I end up waiting in long lines at the counters or dealing with vending machines that are slow, offline, or hard to use.
Because	Ticket issuing is still mostly manual, counters aren't always staffed during busy periods, and fares are worked out by hand.

1.4 Key Points

- Extended waiting lines at counters during office hours, exam periods, and festival seasons.
- Hand-calculated fares take time and are susceptible to mistakes.
- Counters keep restricted hours even though travel demand is round the clock.
- Misplaced or damaged tokens/tickets lead to delays for re-verification.
- Minimal support at manual counters for elderly, disabled, or visually impaired riders.
- No single digital record of a commuter's trip history or spending.
- Running counters, staffing them, and stocking tokens is costly for metro operators.
- Fare collection isn't linked to any workflow-based customer support or refund process.

1.5 Impact of the Problem

Problem statement canvas

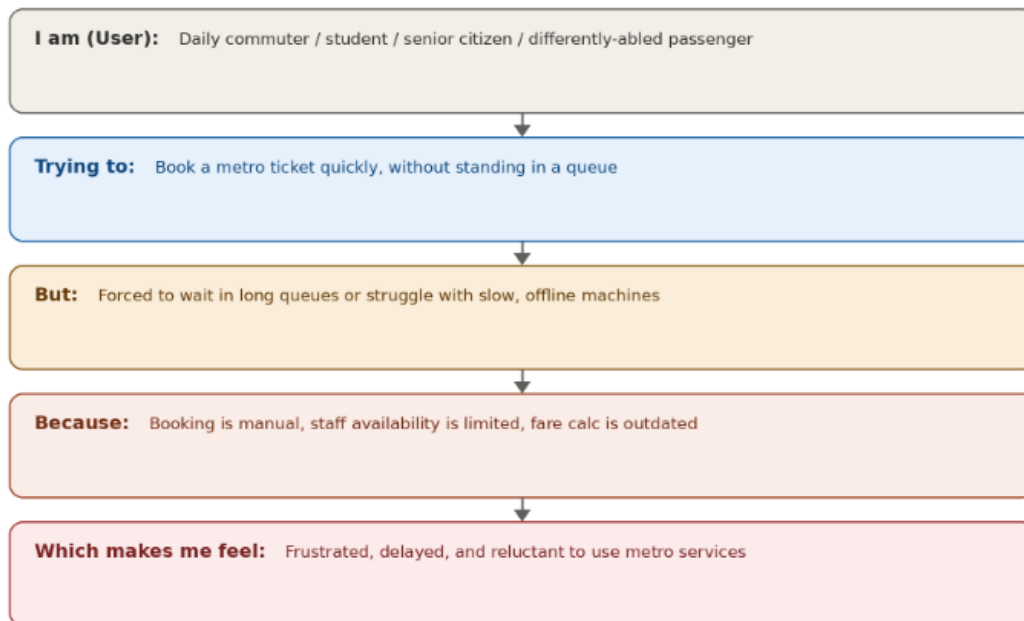


Figure — Consequences of the current manual ticketing process

1.6 Need for a Digital Solution

What's clearly missing is a self-service digital way to book a metro ticket — one where commuters pick their stations, see the fare worked out automatically, and receive a QR-based digital ticket right away, with no need to approach a counter. Building this on ServiceNow means the solution can draw on the Service Catalog for a guided booking journey, Flow Designer for automatic fare computation and ticket issuance, and Notifications to deliver the digital ticket instantly — giving commuters a quicker, more inclusive, and more dependable travel experience.

Impact by stakeholder



Figure — Rationale for a digital, self-service booking solution